Condor: A Neural Connection Network for Enhanced Attention based on Generalized Local Analysis

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Abstract

The attention mechanism in traditional neural networks relies on pairwise interactions between tokens, limiting its ability to capture complex, multi-token relationships. This study introduces Condor, a novel architecture that extends the attention mechanism through a neural connection network based on the Generalized Local Analysis (GLA) framework. Our approach replaces static attention patterns with learnable basis functions that dynamically model relationships within a local window. The Condor architecture achieves linear computational complexity of O(LWH) while maintaining the expressive power to capture sophisticated sequence patterns. Experimental results on wikitext-2 demonstrate improved perplexity and faster convergence compared to the standard Transformer, confirming that each attention head learns unique connection patterns specializing in different aspects of sequence modeling. Code is available at: `https://github.com/Kim-Ai-gpu/Condor`

1 Introduction

The attention mechanism has become the cornerstone of modern neural network architectures, especially in sequence modeling tasks. Since the introduction of the Transformer [29], self-attention has proven highly effective at capturing long-range dependencies and contextual relationships in sequences. However, traditional attention mechanisms are fundamentally constrained by the pairwise interaction paradigm, which calculates attention weights based solely on the similarity between pairs of tokens.

This pairwise constraint manifests in several ways. First, standard attention cannot directly model complex relationships involving multiple tokens simultaneously. Second, the attention pattern is determined by a fixed mathematical operation (typically a dot product followed by a softmax), which limits its adaptability to diverse sequence structures. Third, its quadratic complexity with respect to sequence length, $O(L^2)$, makes it computationally impractical for very long sequences.

Recent studies have proposed various approaches to overcome these limitations. Linear attention attempted to reduce computational complexity but at the cost of expressive power, while sparse attention patterns were limited to specific, predefined patterns. Local attention mechanisms improved computational efficiency but still operated within the fundamental framework of pairwise interactions.

In this paper, we present a completely new approach to address these fundamental limitations. We introduce a mathematical framework called Generalized Local Analysis

(GLA). GLA provides a systematic way to design and apply the most appropriate 'mathematical lens' for analyzing the local behavior of a function or data. It generalizes the classical concept of a derivative, moving from a fixed linear approximation to a flexible, multi-dimensional analysis using arbitrary basis functions.

By applying the principles of GLA to the attention mechanism, we develop GLA-Attention. This allows us to generalize the traditional attention pattern to a "learnable connection function," implemented as a neural network.

The main contributions of this study are as follows:

- Introduction of the Generalized Local Analysis (GLA) framework for analyzing local data patterns.
- Application of GLA to create GLA-Attention, a method for dynamically modeling token relationships through learnable basis functions.
- The design of the Condor architecture, achieving O(LWH) linear complexity.
- Multi-scale representation where each attention head learns a different connection pattern.
- Theoretical proof of the expressiveness and convergence of GLA-Attention.
- Experimental validation of performance improvements on the wikitext-2 dataset.

2 Related Work

This work aims to address fundamental limitations of the attention mechanism by introducing a new paradigm. To contextualize our contributions, we review existing literature across four main themes: (1) enhancing the computational efficiency of attention, (2) increasing the expressive power of attention, (3) sophisticated encoding of positional information, and (4) alternative architectures for sequence modeling that challenge the Transformer's dominance.

2.1 Efficient Attention Mechanisms for Addressing Computational Complexity

The self-attention mechanism in the original Transformer [29] has a computational complexity of $O(L^2)$ with respect to the sequence length L, as it computes interactions for all token pairs. This quadratic complexity has been a major bottleneck for applications involving long sequences, prompting extensive research into more efficient alternatives.

2.1.1 Low-Rank and Kernel-Based Approximations

Early efforts focused on mathematically approximating the attention matrix to achieve linear ($O(L)$) complexity. **Linformer** [30] was based on the observation that the self-attention matrix is often low-rank and proposed projecting the keys and values into a lower-dimensional space to reduce computational cost. **Performer** [4] introduced the FAVOR+ mechanism, which uses kernel functions to approximate the attention mechanism, also achieving linear complexity. In a similar vein, **Nyströmformer** [31] applied

the Nyström method for a low-rank approximation of the attention matrix. Meanwhile, **Reformer** [15] employed Locality-Sensitive Hashing (LSH) to selectively compute attention only for query-key pairs that are likely to be similar. While these methods significantly improve computational efficiency, they carry the potential risk of sacrificing some expressive power due to the approximation.

2.1.2 Sparse and Local Attention Patterns

Another major line of research restricts the scope of attention to a subset of the sequence, creating sparse or local attention patterns. **Longformer** [3] combined a sliding window-based local attention with task-specific global attention on certain tokens, proving effective for long document processing. **BigBird** [32] combined local windowed attention, random attention, and global attention to maintain Turing-completeness while achieving near-linear complexity. **ETC (Extended Transformer Construction)** [1] also utilized a combination of local and global attention. More dynamically, **Routing Transformer** [23] used k-means clustering to group tokens and performed attention only within each cluster. Although these approaches greatly enhance the ability to process long sequences, their flexibility is often limited by pre-defined, static attention patterns. In contrast, Condor learns the interaction patterns themselves from data within a purely local window.

2.1.3 Hardware-Aware Optimization

Separate from algorithmic improvements, significant gains have been achieved by optimizing the attention computation for modern hardware architectures. **FlashAttention** [6] and its successor, **FlashAttention-2** [5], do not change the $O(L^2)$ complexity but re-engineer the attention algorithm to be I/O-aware. By minimizing data movement between the GPU's high-bandwidth memory (HBM) and on-chip SRAM using techniques like tiling and recomputation, they achieve dramatic speedups and memory savings. This has become the de facto standard for training and inference in most large-scale language models (LLMs).

2.2 Beyond Pairwise Interactions: Enhancing Attention's Expressive Power

Standard attention relies on a fixed mechanism based on dot-product similarity between queries and keys. Some research has sought to overcome the limitations of this pairwise interaction to model richer relationships. **Synthesizer** [28] proposed synthetic attention mechanisms that generate attention weights directly from each token without explicit pairwise interactions. **Talking-Heads Attention** [25] introduced additional transformations that allow attention heads to communicate with each other after computing attention weights, enabling synergy between heads. The **Mixture of Experts (MoE)** paradigm [26, 16, 7], while not an attention mechanism itself, shares the philosophy of dynamic, input-dependent computation by routing each token to specialized expert MLPs. Our Generalized Local Analysis (GLA) framework takes this a step further by modeling the relationship between tokens itself as a 'learnable basis function'. Instead of using a fixed similarity function, it learns the most appropriate 'mathematical lens' for the data, attempting a more fundamental generalization to capture complex, multi-token relationships.

2.3 The Evolution of Positional Information: Learnable Positional Representations

The method for modeling token order significantly impacts Transformer performance. Following the initial fixed sinusoidal positional embeddings [29], various techniques have been proposed to represent relative positions more effectively. **Relative Position Bias** [24, 20] adds a learnable scalar bias to the attention score based on the relative distance between the key and query. **DeBERTa** [12] advanced this with a disentangled attention mechanism that computes attention using separate vectors for content and position. More recently, **RoPE (Rotary Position Embedding)** [27] applies absolute positional information to queries and keys via rotation matrices, which naturally incorporates relative position information during the dot-product operation. **ALiBi (Attention with Linear Biases)** [18] demonstrated remarkable extrapolation capabilities for longer sequences by simply adding a fixed, distance-proportional penalty to attention scores. Our Condor architecture provides a more general framework that encompasses these ideas by using positional information as an input to the learnable connection function. This allows it to learn the dynamic 'shape' of the relationship based on position, rather than merely adding a bias or applying a fixed transformation.

2.4 Alternatives to the Transformer: New Sequence Modeling Architectures

Recently, research into fundamental alternatives to the Transformer has gained significant traction, with State Space Models (SSMs) showing particular promise. **S4 (Structured State Space for Sequence Modeling)** [10] reinterpreted classic state space models for continuous signals and utilized the HiPPO framework to effectively capture long-range dependencies, achieving performance competitive with Transformers at linear complexity. Building on S4, **Mamba** [9] introduced a selective mechanism that makes the SSM parameters input-dependent, allowing it to selectively remember or forget information based on context, which led to substantial performance gains. Other research, such as **RWKV** [17], has attempted to combine the efficiency of RNNs with the performance of Transformers to achieve linear scaling. While these models process sequences in a fundamentally different way than attention, they share the common goal of efficiently modeling long-range dependencies. Condor remains within the attention paradigm but introduces a new mathematical tool, GLA, to enhance dynamic local relationship modeling, presenting a novel path to compete with these alternative architectures.

2.5 Positioning of Our Research

In summary, previous work has largely focused on improving specific aspects of attention, such as efficiency, sparse patterns, or positional representations. While our work, Condor, builds upon these insights, it distinguishes itself by introducing the Generalized Local Analysis (GLA) framework. This framework aims for a more fundamental level of generalization and expressive power by replacing the core token-to-token interaction with a 'learnable, dynamic function', thereby creating a more powerful and flexible attention mechanism.

3 Generalized Local Analysis (GLA) Framework

3.1 Philosophy and Goal

GLA is not an attempt to replace or generalize calculus. GLA is a mathematical framework for analyzing the local behavior of a function or time-series data. Its core objective is to systematically design and apply the most suitable 'mathematical lens' for the problem at hand, thereby extracting multi-dimensional information that cannot be found with a single, conventional perspective (i.e., linear approximation).

3.2 Core Components

GLA is defined by the combination of three core components: (f, g, R).

- **Target Function,** f: The continuous function or discrete time-series data to be analyzed. $f : \mathbb{R} \to \mathbb{R}$.
- **Basis Function,** g: A parameterized function selected by the analyst to model the local patterns of the function. $g : \mathbb{R} \times \Theta \to \mathbb{R}$, where $\theta \in \Theta \subset \mathbb{R}^n$ is an n-dimensional parameter vector.
- **Basis Dictionary,** D: To prevent arbitrary selection of g, we can define a set of predefined, useful basis functions, $D = \{g_{\text{lin}}, g_{\text{poly}}, g_{\text{exp}}, g_{\text{osc}}, ...\}$. Each basis function is designed to capture a specific type of dynamic (e.g., linear change, accelerated motion, exponential growth, oscillation).
- **Selection Principle,** R: A mathematical principle for selecting a single optimal parameter $\theta^*(h)$ from the countless basis functions $g(t; \theta)$ that pass through two given points $(x_0, f(x_0))$ and $(x_0 + h, f(x_0 + h))$. R is not an arbitrary constraint but an explicit statement of the analyst's prior assumption about "which solution is most desirable."
  - $R_{\textbf{parsimony}}$ **(Minimum Parameter Principle / L2-norm)**: Selects the 'simplest' or 'lowest energy' solution. Geometrically, it prefers the smoothest connection. $\min \|\theta\|_2$.
  - $R_{\textbf{sparsity}}$ **(Sparsity Principle / L1-norm)**: Selects the solution described by the fewest non-zero parameters. Useful for eliminating unnecessary complexity. $\min \|\theta\|_1$.
  - $R_{\textbf{bayesian}}$ **(Bayesian Principle)**: Defines a prior probability distribution $P(\theta)$ for the parameters and selects the solution that maximizes the posterior probability. This most actively incorporates domain knowledge.

3.3 Core Definition: Generalized Local Derivative (GLD)

Definition 3.1 (Generalized Local Derivative, GLD)**.** *The Generalized Local Derivative (GLD) of a function f at point x_0, with respect to a basis function g and a selection principle R, denoted as $\nabla[f; g, R](x_0)$, is defined as follows:*

1. *For any $h \neq 0$, define the constraint set $C(h) = \{\theta | g(x_0; \theta) = f(x_0), g(x_0 + h; \theta) = f(x_0 + h)\}$. Let $S(h)$ be the set of parameters satisfying these constraints.*

2. *Apply the selection principle R to $S(h)$ to find the unique optimal parameter: $\theta^*(h) = \arg\min_{\theta \in S(h)} R(\theta)$.*

3. *If the limit of this optimal parameter exists, it is defined as the GLD.*

$$\nabla[f; g, R](x_0) := \lim_{h \to 0} \theta^*(h) \tag{1}$$

This GLD is not an absolute property of f, but rather an n-dimensional information vector obtained by observing f at point x_0 through the lens of g and the philosophy of R.

3.4 Key Theoretical Results

This framework is powerful because the following theorems can be proven.

Theorem 3.2 (Consistency with Classical Calculus)**.** *Let the basis function be linear, $g_{lin}(t; a, b) = a(t - x_0) + b$. For any selection principle R that ensures the uniqueness of $\theta^*(h) = (a(h), b(h))$, the first component of the GLD is equal to the classical derivative.*

$$[\nabla[f; g_{lin}, R](x_0)]_1 = f'(x_0) \tag{2}$$

(Proof: This shows that GLA is an extension that naturally includes classical calculus.)

Theorem 3.3 (Existence and Uniqueness Condition)**.** *If the function f is C^n (n-times differentiable) at x_0, the basis function g has sufficiently nice properties with respect to its parameters θ (e.g., continuously differentiable, Jacobian matrix has full rank), and the selection principle R defines a convex optimization problem, then the GLD $\nabla[f; g, R](x_0)$ exists and is unique. (Proof: This ensures that GLA is a mathematically well-defined operation.)*

Theorem 3.4 (Foundation for Multi-stage Analysis - Parameter Signals)**.** *Calculating the GLD $\nabla[f; g, R](x)$ for all x results in a new set of functions representing the trajectory of the k-th parameter, known as Parameter Signals: $\theta_k(x) = [\nabla[f; g, R](x)]_k$. These parameter signals can themselves become new subjects for analysis.*

3.5 Illustrative Application: Predicting Fatigue Failure in Materials Engineering

The practical utility of a theory is proven by its ability to solve concrete problems.

Problem: When a metal material is subjected to repeated stress, micro-cracks accumulate internally, leading to sudden failure. Early prediction of this failure point is crucial.

Limitation of Existing Methods: The stress-strain curve of a material exhibits nearly linear elastic behavior until just before fracture. The first derivative (elastic modulus) remains almost constant, making prediction difficult.

GLA Approach:

1. **Goal**: Detect the minute non-linear 'fatigue' effects that deviate from linearity.

2. **Basis Function Selection (g)**: Based on materials science models, we select a basis function that adds a non-linear 'damage' term to the ideal linear elastic model: $g_{\text{damage}}(\epsilon; E, \beta) = E \cdot \epsilon - \beta \cdot \epsilon^3$. Here, ϵ is the strain, E is the elastic modulus, and β is the **'non-linear damage coefficient'**.

3. **Selection Principle Selection (R)**: To find the most physically stable solution, we choose $R_{\text{parsimony}}$ (L2-norm).

4. **Calculation**: During a stress test, compute the GLD on the stress-strain curve $\sigma(\epsilon)$: $\nabla[\sigma; g_{\text{damage}}, R_{\text{parsimony}}](\epsilon) = (E(\epsilon), \beta(\epsilon))$.

Results and Interpretation:

- The $E(\epsilon)$ parameter signal (elastic modulus) will remain nearly constant until just before fracture.
- The $\boldsymbol{\beta(\epsilon)}$ parameter signal (damage coefficient) will initially be close to zero, but will show a pattern of gradual increase to a significant positive value as micro-cracks begin to accumulate inside the material.

Conclusion: The moment $\beta(\epsilon)$ exceeds a certain threshold can be defined as a precursor to failure. This demonstrates how GLA can 'design' and extract a new indicator for the 'health status' of the material—an indicator that is invisible when looking only at $\sigma'(\epsilon)$. This is how GLA combines domain knowledge with a mathematical framework to create new, problem-solving information.

4 Methodology

4.1 Applying GLA to Token Sequences

We can apply the GLA framework by interpreting a token sequence as a discretely sampled function. If we interpret the sequence $[token_1, token_2, \ldots, token_n]$ as a function $f(x)$, then $f(1) = token_1, f(2) = token_2, \ldots$.

The window $[token_{i-k}, token_{i-k+1}, \ldots, token_{i+k}]$ can be seen as a local sampling of the function. Within the GLA framework, our learnable attention mechanism becomes a specific choice of a basis function $g(t; \theta)$ connecting these points. The parameters θ of this basis function, which represent the "dynamic characteristics" of that interval, are learned by the neural network.

By physical intuition, if we apply a parabolic basis function $g(t) = a(t - t_0)^2 + b(t - t_0) + c$ to a position function $s(t)$, the resulting GLD parameters (a, b, c) correspond to $(\text{acceleration}/2, \text{velocity}, \text{position})$, respectively. Similarly, in a token sequence, we can extract semantic dynamics such as the rate of grammatical change, the strength of semantic connections, and the contextual baseline.

4.2 GLA-Attention

GLA-Attention extends traditional self-attention to dynamically model complex relationships between tokens through a learnable basis function. Each attention head has an independent basis function network, allowing it to learn different kinds of relationships.

4.2.1 Connection Network (Basis Function) Structure

The basis function for each attention head is implemented as the following neural network:

$$g_h(t) = \text{SwiGLU}h(t) \tag{3}$$

$$= W\text{gate}^{(h)} \cdot t \odot \text{Swish}(W_{\text{up}}^{(h)} \cdot t + b_{\text{up}}^{(h)}) \cdot W_{\text{down}}^{(h)} + b_{\text{out}}^{(h)} \tag{4}$$

where $t \in [0, 1]$ is the normalized positional information, and h is the head index.

4.2.2 Algorithm Steps

The computation process of GLA-Attention is as follows:

Step 1: Standard Attention Calculation For an input sequence $X \in \mathbb{R}^{L \times d}$:

$$Q = XW^Q, \quad K = XW^K, \quad V = XW^V \tag{5}$$

$$Q, K, V \in \mathbb{R}^{L \times d} \to \mathbb{R}^{B \times H \times L \times d_h} \tag{6}$$

Step 2: Local Window Generation For a window size W, add padding and create a sliding window:

$$K_{\text{pad}} = \text{Pad}(K, W/2) \tag{7}$$

$$K_{\text{win}} = \text{Unfold}(K_{\text{pad}}, W) \in \mathbb{R}^{B \times H \times L \times W \times d_h} \tag{8}$$

The same is applied to V.

Step 3: Base Attention Scores Attention scores with tokens within the window at each position i:

$$\text{scores}_{i,j} = \frac{Q_i \cdot K_{\text{win}_{i,j}}}{\sqrt{d_h}} \tag{9}$$

Softmax normalization:

$$\alpha_{i,j} = \frac{\exp(\text{scores}_{i,j})}{\sum_{k=1}^{W} \exp(\text{scores}_{i,k})} \tag{10}$$

Step 4: Applying the Basis Function (Connection Function) Connection weights for relative positions within the window:

$$t_j = \frac{j-1}{W-1}, \quad j = 1, 2, \ldots, W \tag{11}$$

$$c_{h,j} = g_h(t_j) = \text{MLP}_h(t_j) \tag{12}$$

$$\beta_{h,j} = \frac{\exp(c_{h,j})}{\sum_{k=1}^{W} \exp(c_{h,k})} \tag{13}$$

Step 5: Final Weight Combination Combination of attention scores and connection weights:

$$w_{i,j}^{(h)} = \alpha_{i,j} \cdot \beta_{h,j} \tag{14}$$

Normalization:

$$\tilde{w}_{i,j}^{(h)} = \frac{w_{i,j}^{(h)}}{\sum_{k=1}^{W} w_{i,k}^{(h)} + \epsilon} \tag{15}$$

Step 6: Output Calculation Final output:

$$\text{Output}_i^{(h)} = \sum_{j=1}^{W} \tilde{w}_{i,j}^{(h)} \cdot V_{\text{win}_{i,j}} \tag{16}$$

5 Theoretical Analysis

5.1 Expressive Power Analysis

Theorem 5.1 (Inclusion of Standard Attention)**.** *GLA-Attention includes Standard Self-Attention as a special case.*

Proof. If we set the basis function $g_h(t) = c$ (a constant function):

1. Connection weights: $\beta_{h,j} = \frac{\exp(c)}{\sum_{k=1}^{W} \exp(c)} = \frac{1}{W}$ (uniform)
2. Final weights: $w_{i,j}^{(h)} = \alpha_{i,j} \cdot \frac{1}{W}$
3. Output: $\text{Output}_i^{(h)} = \frac{1}{W} \sum_{j=1}^{W} \alpha_{i,j} \cdot V_{\text{win}_{i,j}}$

In the limit where $W \to L$ (the entire sequence) and without padding:

$$\text{Output}_i^{(h)} = \sum_{j=1}^{L} \alpha_{i,j} \cdot V_j$$

This is identical to standard self-attention. □

Theorem 5.2 (Expanded Expressive Power)**.** *GLA-Attention can express richer relationships than Standard Attention.*

Proof. For the class of basis functions $\mathcal{G} = \{g : [0, 1] \to \mathbb{R} | g \text{ is a learnable function}\}$, the uniform weighting of Standard Attention is a special case where $g(t) = c$.

For any non-constant basis function $g^* \in \mathcal{G}$, we can generate position-specific weights as follows:

$$\beta_{h,j}^* = \frac{\exp(g^*(t_j))}{\sum_{k=1}^{W} \exp(g^*(t_k))} \neq \frac{1}{W}$$

This provides adaptive weighting based on position, modeling complex positional dependencies that Standard Attention cannot express. □

Theorem 5.3 (Universal Approximation)**.** *Given a sufficient number of heads and basis function capacity, GLA-Attention can approximate any attention function.*

Proof. Let's define the function space as:

$$\mathcal{F} = \{f : \mathbb{R}^{L \times d} \to \mathbb{R}^{L \times d} | f \text{ is an attention transformation}\}$$

By the Universal Approximation Theorem [13], a sufficiently wide MLP can approximate any continuous function to an arbitrary precision.

For any target attention function $f^* \in \mathcal{F}$ and a given $\epsilon > 0$:

Step 1: Implement the basis function g_h of each head h as an MLP with sufficient capacity.

Step 2: There exist a sufficient number of heads H and appropriate combination weights λ_h such that:

$$\sup_{X \in \mathcal{X}} \left\| \sum_{h=1}^{H} \lambda_h \cdot \text{GLA-Attn}_h(X) - f^*(X) \right\|_F < \epsilon$$

Step 3: If we bound the MLP approximation error of each g_h by δ, the total approximation error is proportional to $H \cdot \delta$. Thus, by selecting an appropriate H and MLP capacity, any ϵ can be achieved. □

5.2 Complexity Analysis

Theorem 5.4 (Time Complexity)**.** *The time complexity of GLA-Attention is* $O(LWH)$*.*

Proof. Analyzing the complexity of each step:

Preprocessing Stage:

- Query, Key, Value calculation: $O(Ld^2H)$
- Window generation (Padding + Unfold): $O(LW \cdot d)$

Core Operations:

- Attention score calculation: Dot product with W tokens in the window for each position i.
$$O(L \cdot W \cdot d_h \cdot H) = O(LWd) \text{ (since } d_h = d/H)$$
- Basis function calculation: MLP operation for each head.
$$O(H \cdot W \cdot C) = O(HW) \text{ (where } C \text{ is MLP operation constant)}$$
- Final weight calculation and output: $O(LWH)$

Excluding constant terms, the dominant term is $O(LWH)$. Compared to the $O(L^2H)$ of Standard Attention, this is a significant improvement when $W \ll L$. □

Theorem 5.5 (Space Complexity)**.** *The space complexity of GLA-Attention is* $O(LWd)$*.*

Proof. Analyzing the main memory usage:

$$\text{Input storage} : O(Ld) \tag{17}$$
$$\text{Window tensor} : O(L \cdot W \cdot H \cdot d_h) = O(LWd) \tag{18}$$
$$\text{Attention scores} : O(LWH) \tag{19}$$
$$\text{Connection weights} : O(WH) \tag{20}$$
$$\text{Intermediate results} : O(LWH) \tag{21}$$
$$\text{Output} : O(Ld) \tag{22}$$

The total space complexity is $O(\max(Ld, LWd, LWH, WH)) = O(LWd)$ since typically $d \gg H$ in transformer architectures, making the window tensor storage the dominant term.

For attention computation specifically, the space requirement is $O(LWH)$ compared to $O(L^2H)$ in Standard Attention, providing significant memory savings when $W \ll L$. However, the overall space complexity is dominated by the $O(LWd)$ term from storing windowed key-value pairs. □

5.3 Convergence Analysis

Theorem 5.6 (Stability of Basis Function Learning)**.** *Under appropriate initialization and regularization conditions, the learning of the basis function converges stably.*

Proof. Let's assume the basis function $g_h : [0,1] \to \mathbb{R}$ is L-Lipschitz continuous:

$$|g_h(t_1) - g_h(t_2)| \leq L|t_1 - t_2|, \quad \forall t_1, t_2 \in [0,1]$$

Gradient Bound: The gradient of the loss function $\mathcal{L}$ with respect to the basis function is:

$$\left\| \frac{\partial \mathcal{L}}{\partial g_h} \right\| \leq C \cdot L \cdot \|X\|_F \cdot \|\nabla_{\text{output}} \mathcal{L}\|$$

where C is the Lipschitz constant of the softmax function.
Stability Conditions:

1. **Initialization**: Xavier/He [8, 11] initialization ensures $\mathbb{E}[\|g_h(0)\|^2] = O(1)$.
2. **Normalization**: Applying Layer Normalization [2] reduces internal covariate shift.
3. **Gradient Clipping**: $\|\nabla g_h\| \leq \tau$ (for a threshold τ).
4. **Learning Rate Scheduling**: Adaptive learning rate adjustment.

Under these conditions, the SGD update satisfies:

$$\mathbb{E}[\|\theta_{t+1} - \theta^*\|^2] \leq (1 - \mu\eta)\mathbb{E}[\|\theta_t - \theta^*\|^2] + \sigma^2\eta^2$$

where μ is the strong convexity constant and σ^2 is the variance of the gradient noise. Convergence is guaranteed by the Robbins-Siegmund theorem [22]. □

Theorem 5.7 (Gradient Propagation Stability)**.** *GLA-Attention mitigates the vanishing/exploding gradient problem compared to the conventional Transformer.*

Proof. **Locality Property**: Due to the window-based operation:

$$\frac{\partial \text{Output}_j}{\partial X_i} = \begin{cases} \text{non-zero} & \text{if } |i-j| \leq W/2 \\ 0 & \text{otherwise} \end{cases}$$

Gradient Bound:

$$\left\| \frac{\partial \mathcal{L}}{\partial X_i} \right\| \leq \sum_{|j-i| \leq W/2} \left\| \frac{\partial \mathcal{L}}{\partial \text{Output}_j} \right\| \cdot \left\| \frac{\partial \text{Output}_j}{\partial X_i} \right\|$$

Due to the softmax normalization of the basis function $\sum_{j=1}^{W} \beta_{h,j} = 1$:

$$\left\| \frac{\partial \text{Output}_j}{\partial X_i} \right\| \leq M_{\text{softmax}} \cdot M_{\text{basis}}$$

where $M_{\text{softmax}} \leq 1$, and M_{basis} is the maximum slope of the basis function.
Therefore, when $W \ll L$ and the basis function is appropriately regularized:

$$\left\| \frac{\partial \mathcal{L}}{\partial X_i} \right\| \leq O(W) \cdot M_{\text{basis}} \cdot \left\| \frac{\partial \mathcal{L}}{\partial \text{Output}} \right\|$$

This is much more stable than the $O(L)$ dependency in Standard Attention. □

5.4 Multi-Scale Representation Learning

Each head in GLA-Attention learns a different connection pattern to achieve a multi-layered understanding of the sequence. For example, heads could potentially specialize as follows:

$$\text{Head } h_1: \quad \nabla[\text{sequence}; g_1, R] \to \text{Short-range grammatical patterns} \tag{23}$$
$$\text{Head } h_2: \quad \nabla[\text{sequence}; g_2, R] \to \text{Mid-range semantic associations} \tag{24}$$
$$\text{Head } h_3: \quad \nabla[\text{sequence}; g_3, R] \to \text{Long-range discourse structure} \tag{25}$$
$$\text{Head } h_4: \quad \nabla[\text{sequence}; g_4, R] \to \text{Thematic consistency} \tag{26}$$

(Where R is the implicit selection principle of neural network optimization) These examples illustrate the potential ways in which heads could plausibly differentiate themselves, though the actual learned patterns would be determined empirically through training.

Theorem 5.8 (Multi-Pattern Decomposability)**.** *A GLA-Attention with H heads can decompose an input sequence into H different perspectives.*

Proof. Let's assume the basis function g_h of each head h is learned from a different function class:

$$g_h \in \mathcal{G}_h, \quad \mathcal{G}_i \cap \mathcal{G}_j = \emptyset \text{ for } i \neq j$$

The overall representation for a sequence S is:

$$\text{Repr}(S) = \bigoplus_{h=1}^{H} \text{GLA-Transform}(S; g_h)$$

where $\bigoplus$ is the concatenation or combination operation of the head-wise representations. Since each g_h is learned independently, the following holds:

$$\text{Mutual Information}(\text{GLA-Transform}(S; g_i), \text{GLA-Transform}(S; g_j)) \to \min$$

This implies that each head extracts different, mutually complementary patterns. □

6 Experiments

6.1 Experimental Setup

We evaluated the performance of the GLA-Attention architecture across a variety of language modeling tasks to assess its effectiveness and characteristics. Due to computational resource constraints, our experiments were conducted on relatively small-scale models to provide initial validation of the proposed approach. Unless specified otherwise, experiments were conducted on the Wikitext-2 dataset. The primary hyperparameters used in our framework are as follows:

- Model dimension: $d_{\text{model}} = 256$
- Number of layers: $L = 4$
- Number of attention heads: $H = 4$

- Feed-forward dimension multiplier: 4
- Maximum sequence length: 256
- Window size: $W = 32$ (for local and GLA-Attention methods)
- Batch size: 16
- Training epochs: 3
- Optimizer: AdamW [14]
- Learning rate: 5×10^{-4}
- Weight decay: 0.01
- Learning rate scheduler: Cosine Annealing

All experiments were performed in a CUDA-supported GPU environment. We used the GPT-2 tokenizer [19] for text processing. The 'Baseline' model in our comparisons refers to a standard Llama architecture using full (non-windowed) self-attention.

6.2 Component Isolation Analysis

To precisely measure the contribution of each element within GLA-Attention, we conducted a component isolation experiment. We compared three distinct models: (1) a Baseline model with standard self-attention, (2) a model using only Local Windowed Attention, and (3) our proposed GLA-Attention.

Table 1: Component Isolation results on Wikitext-2, showing the progressive improvement of each component.

Model	**Validation Loss**	**Perplexity**
Baseline (Standard Attention)	4.8202	124.00
Local Window Attention	4.8059	122.22
GLA-Attention	**4.7624**	**117.03**

As shown in Table 1, simply applying a local window provides a marginal improvement over the baseline. However, the introduction of the neural connection weights in GLA-Attention leads to a significant reduction in both validation loss and perplexity. This result highlights that the dynamic, position-aware weighting mechanism is the key factor driving the performance gains.

6.3 Comparison with Efficient Attention Mechanisms

We performed a fair comparison between GLA-Attention and other popular efficient attention mechanisms to position its performance within the current landscape. Note that the Longformer and BigBird implementations are simplified versions designed to capture their core ideas for this comparative study.

The results in Table 2 indicate that GLA-Attention significantly outperforms standard Local Attention and Linformer. While the simplified Longformer and BigBird models achieved the best performance, GLA-Attention demonstrates competitive results without relying on global tokens or complex, pre-defined sparse patterns. Its strength lies in enhancing a purely local window with dynamic, learned weights.

Table 2: Performance comparison against other attention mechanisms on Wikitext-2.

Model	Validation Loss	Perplexity
Local Attention	4.8033	121.91
Linformer	4.8320	125.46
GLA-Attention	4.7652	117.36
BigBird (Simplified)	4.6657	106.24
Longformer (Simplified)	**4.6530**	**104.89**

6.4 Ablation Studies

6.4.1 Impact of Window Size

To analyze the sensitivity of GLA-Attention to its primary hyperparameter, the window size (W), we conducted an ablation study with varying sizes. We tested window sizes of 8, 16, and 32.

Table 3: Ablation study on the window size (W) for GLA-Attention on Wikitext-2.

Window Size	Validation Loss	Perplexity
W = 8	4.7528	115.91
W = 16	**4.7494**	**115.51**
W = 32	4.7637	117.18

As shown in Table 3, a window size of 16 delivered optimal performance on this task. A smaller window of 8 was slightly less effective, likely due to a more constrained receptive field. Interestingly, increasing the window size to 32 slightly degraded performance, suggesting that the fixed-size neural network may struggle to generalize its learned patterns over a wider context or that the additional tokens do not contribute effectively. This highlights a crucial trade-off in selecting an appropriate window size.

6.5 Generalization Across Diverse Datasets

To evaluate the robustness and versatility of GLA-Attention, we benchmarked it against the baseline on three datasets with distinct characteristics: `gsm8k` (mathematical reasoning), `code search net` (Python code and documentation), and a dataset of arXiv papers (complex scientific language).

Table 4: Performance comparison on diverse datasets, measured by perplexity.

Dataset	Model	Final Perplexity	Improvement
gsm8k	Baseline	48.53	10.47%
	GLA-Attention	**43.45**	
code search net	Baseline	34.66	11.48%
	GLA-Attention	**30.68**	
arXiv Papers	Baseline	293.33	8.99%
	GLA-Attention	**266.95**	

The results in Table 4 show that GLA-Attention consistently outperformed the standard attention baseline across all tasks, achieving perplexity reductions of **10.47%** on

`gsm8k`, **11.48%** on `code search net`, and **8.99%** on arXiv papers. This demonstrates its ability to effectively model various data types, from structured code to mathematical problems and dense scientific text. The consistent improvements suggest that its dynamic weighting mechanism is a broadly applicable enhancement. As expected, the high perplexity values on the arXiv dataset reflect its significant complexity and large vocabulary compared to the other corpora.

7 Conclusion and Future Work

In this study, we proposed GLA-Attention, a novel attention mechanism based on the Generalized Local Analysis (GLA) framework. This approach overcomes the limitations of traditional pairwise interactions and dynamically models complex relationships between tokens through learnable basis functions.

The main achievements are as follows:

- Successful application of the GLA framework to the attention mechanism.
- Improved computational efficiency by achieving linear complexity $O(L \times W)$.
- Superior performance and faster convergence compared to the standard Transformer.
- Confirmation of specialized connection pattern learning in multi-head attention.
- Theoretical guarantees of expressiveness and convergence.

Future research directions include:

- Exploring various basis function architectures (CNN, RNN, Attention-based).
- An adaptive window size adjustment mechanism.
- Extension and application to other domains (vision, speech, materials science).
- Scalability research for longer sequences.
- Multi-scale modeling through hierarchical GLA.
- Methods to improve the interpretability of learned basis functions.

In particular, the mathematical foundation of the GLA framework is applicable to other neural network components, holding the potential to create new variations of recurrent neural networks, convolutional neural networks, and more. It is also significant in terms of interpretability, as the learned patterns of the basis function can help us better understand the model's decision-making process.

A Appendix

A.1 Application Possibilities

A.1.1 Application to Computer Vision

Applying the GLA framework to sequences of image patches:

- **Spatial Basis Function**: Modeling relationships between patches in 2D space.
- **Scale Basis Function**: Learning relationships between different resolutions.
- **Channel Basis Function**: Modeling interactions between color/feature channels.

A.1.2 Application to Speech Processing

Applying GLA to audio sequences:

- **Temporal Basis Function**: Modeling the temporal dynamics of speech.
- **Frequency Basis Function**: Relationships between frequencies in a spectrogram.
- **Prosodic Basis Function**: Learning patterns of intonation, stress, and rhythm.

A.1.3 Application to Graph Neural Networks

Applying GLA to graph-structured data:

- **Path Basis Function**: Modeling the functional relationship of paths between nodes.
- **Neighborhood Basis Function**: Learning the dynamics of local neighborhood structures.
- **Hierarchical Basis Function**: Capturing multi-scale graph patterns.

B Detailed Comparison with Related Work

B.1 Comparison with Existing Efficient Attention Methods

Table 5: Comparison of Attention Methodologies

Method	Complexity	Expressiveness	Interpretability	Adaptability
Standard Attention	$O(L^2)$	High	Medium	Low
Linear Attention	$O(L)$	Low	Low	Low
Sparse Attention	$O(L\sqrt{L})$	Medium	Low	Low
Local Attention	$O(LW)$	Medium	Medium	Low
GLA-Attention	$O(LWH)$	High	High	High

B.2 Relationship with Positional Encoding Methods

The basis function of GLA-Attention has the following relationship with existing positional encoding methods:

- **Absolute PE**: Fixed positional information vs. learnable connection relationships.
- **Relative PE**: Based on relative distance vs. based on functional relationships.
- **RoPE**: Rotational transformation vs. general connection transformation.
- **ALiBi**: Linear decay vs. learnable arbitrary function.

C Limitations and Future Research Directions

C.1 Current Limitations

- **Hyperparameter Sensitivity**: The choice of window size and basis function structure has a large impact on performance.
- **Interpretation Complexity**: Interpreting the meaning of the learned basis functions is still challenging.
- **Very Long Sequences**: Currently optimized for medium-length sequences.
- **Domain Specificity**: A need to design basis functions suitable for each domain.

C.2 Future Research Directions

C.2.1 Automated Architecture Search

Applying Neural Architecture Search (NAS) [33, 21] to automatically search for the optimal structure of the basis function:

- Evolutionary algorithm-based search for basis function structure.
- Reinforcement learning for adaptive window size adjustment.
- Automatic discovery of optimal connection patterns for specific tasks.

C.2.2 Theoretical Extensions

- Information-theoretic analysis of the GLA framework.
- PAC-Bayes analysis of the expressive power and generalization ability of basis functions.
- Approximation theory research on various classes of basis functions.

C.2.3 Practical Applications

- Application to large-scale language models and study of scaling laws.
- Extension of cross-attention in multimodal models.
- Hardware optimization for real-time inference.

This research presents a new paradigm for the attention mechanism through the GLA framework, which is expected to provide an important theoretical basis for the future development of neural network architectures. In particular, the concept of a learnable basis function is a universal idea with great potential, applicable not only to attention but also to various other neural network components.

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